Lattice AI at ArGuard Shared Tasks: Encoder Ensembles and a Re-Annotated Test Slice in Arabic Harmful-Prompt Detection

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Abstract

We describe our system for the ArGuard 2026 shared task (Task B), harmful-prompt detection for Arabic, with subtasks B1 (binary safe versus unsafe) and B2 (seven-way harm-domain classification), both scored by macro-F1. Our system pairs a Unicode-normalization front-end with an ensemble of fine-tuned Arabic encoders, treats B2 as topic-domain classification over all prompts, and adds a retrieval-augmented resolver for the confusable interpersonal classes; it reaches development macro-F1 0.907 (B2) and 0.950 (B1), and on the blind test **B1** 0.9839 **(1st, tied)** and **B2** 0.7875 **(2nd)**. Our main contribution explains the large B2 development-to-test gap. One fifth of the B2 test is a "novel" slice: AraSafe prompts re-annotated into the seven ArGuard domains under a fixed cross-taxonomy mapping that the organizers confirmed (Table 1). Because that mapping does not preserve the training convention and is recorded in no data available during the task, no system can infer it, and the competitive B2 field compresses onto a blind 0.68–0.79 band, far below the ≈0.90 reachable in-distribution. On the familiar slice we score macro-F1 0.908, matching development; we argue this is the honest measure of the system.

1 Introduction

Detecting harmful prompts in Arabic compounds several difficulties: dialectal diversity, rich morphology, implicit intent, overlapping harm domains, and intra-script obfuscation. Task B (Alam et al., 2026) asks systems to make a binary safe/unsafe decision (B1) and to assign one of seven harm domains (B2), both under macro-F1.

Our system is deliberately simple and its in-distribution gains come from a few validated decisions (Section 4). But the result we believe is most useful to the community is not the leaderboard number; it is what the blind test revealed about the benchmark. Our B2 system scores 0.907 on development yet 0.7875 on the test, and the entire gap is one 20% slice of the test. We trace that slice to a specific construction: the organizers took prompts verbatim from the public AraSafe corpus (Mubarak et al., 2025) and re-annotated them into the seven ArGuard domains. We confirmed the construction with the organizers: the re-annotated labels follow a fixed mapping from AraSafe's harm taxonomy to the ArGuard label set (Table 1) that does not preserve the training convention's label semantics (Section 6). Predicting these labels therefore means reproducing a mapping recorded in no training, development, or external data available during the task, which is why the slice behaves as an out-of-distribution target for every participant.

Our contributions are: (i) a reproducible Task B system (test B1 0.9839, 1st tied; B2 0.7875, 2nd) whose main lever is treating B2 as topic-domain classification over *all* prompts, safe ones included, worth +0.086 macro-F1; (ii) a complete reverse-engineering of the benchmark's out-of-distribution split; (iii) a documentation, reproducible from the released gold and confirmed by the organizers, of the fixed AraSafe-to-ArGuard mapping that defines the novel slice and how it diverges from the training convention; (iv) model-only diagnostics, a convention-fidelity anti-correlation and agreement-on-the-line over-prediction, that flag the slice as re-annotated before any gold is seen, plus an account of how a convention no participant could infer collapses the whole field onto one band; and (v) a responsibly disclosed identifier-and-template leakage audit that we did not exploit.

2 Related Work

Arabic safety resources and encoders. Task B is built on ArabicDialectSafety (Zaghouani et al., 2026), a dialect-aware Arabic safety benchmark, and reuses prompts from AraSafe (Mubarak et al., 2025). Our models are the standard Arabic

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transformers, MARBERTv2 (Abdul-Mageed et al., 2021), AraBERT (Antoun et al., 2020), CAMeLBERT (Inoue et al., 2021), and AraELECTRA (Antoun et al., 2021), together with the multilingual mmBERT (Marone et al., 2025), and we augment weak classes with back-translation (Sennrich et al., 2016).

Distribution and annotation shift. Our diagnosis of the novel slice draws on methods that predict out-of-distribution accuracy from unlabeled data, agreement-on-the-line (Baek et al., 2022) and average-threshold confidence (Garg et al., 2022), and on the theory of domain divergence (Ben-David et al., 2010, 2006). The signature we observe, capable models scoring *lower* on the slice, is the documented footprint of relabeled data (Sanyal et al., 2024) and of items with negative discrimination in item-response theory (Lalor et al., 2016), while the residual familiar errors reflect intrinsic label uncertainty (Northcutt et al., 2021). Unlike hate-speech generalization studies that shift the data distribution (Yin and Zubiaga, 2021; Röttger et al., 2021), the ArGuard novel slice holds the content fixed and changes only the labeling convention, so it isolates annotation shift from covariate shift.

3 Data and Preprocessing

Task B provides Arabic prompts with a binary label for B1 and a granular harm label for B2 over {adult_content, harm-to-others, fraud, self-harm, bullying, harassment, hate_speech}; the metric is macro-F1. The train (≈16.8k prompts) and development (2,405) sets, each carrying a granular label, are drawn entirely from the ArabicDialectSafety benchmark (Zaghouani et al., 2026), a dialect-aware Arabic safety dataset released by the task organizers; the AraSafe re-annotation described below was applied only when constructing part of the test set. A property that matters for both modeling and analysis is that *safe prompts also carry a domain label*: 3,871 safe training prompts have one of the seven domains (a yoga-and-meditation question is labeled self-harm; a healthy-relationships question is labeled adult_content). B2 is therefore topic-domain classification, not attack detection, and we train the domain classifier on all rows accordingly.

The two test slices. Using only the organizers' released files, an exact-text join between the B1 and B2 test sets partitions the 5,979-row B2 test cleanly into 4,809 *familiar* rows (also present in the B1 test) and 1,170 *novel* rows (absent from it). This structural split, and every decision our submitted system makes, needs no external corpus; the AraSafe provenance below is established only in post-hoc analysis and is never used to label test rows. The familiar rows are drawn from the same ArabicDialectSafety source as train and development (six dialects; 48% Modern Standard Arabic; zero overlap with AraSafe). The novel rows are the opposite on every axis: 100% from AraSafe, 100% Modern Standard Arabic, and 0% present in training. The B1 test contains a disjoint 3,000-row novel block, identified structurally in the same way; predicting this block *safe* scores highly on B1. The block's coincidence with AraSafe's safe half is a post-hoc finding and plays no role in producing the submission. Preprocessing (`normalize_defense`) applies NFKC normalization, strips bidirectional/zero-width controls, elongation, and diacritics, and unifies alef/ya/hamza (Appendix A).

How the novel slice is labeled. The novel rows are AraSafe prompts re-annotated into the ArGuard label set by a fixed mapping from AraSafe's harm categories (Table 1), which the organizers confirmed on request.[1] Terrorism-or-extremism prompts are excluded. This mapping is not label-preserving with respect to the training data: Explicit Content is routed to harassment (not adult_content), Illegal Activities to Bully, and no source category maps to adult_content, which therefore receives zero novel examples although it is the largest class on the familiar slice. Because the mapping appears in no training, development, or external resource available during the task, the novel labels are a deterministic function of a hidden source category and cannot be inferred from the data participants had.

4 System Architecture and Training

All-rows domain training. Because safe prompts carry domain labels and the evaluation mixes safe and unsafe prompts, we train the seven-way classifier on all rows. In a controlled ablation this is worth +0.086 macro-F1 when domains are scored over every prompt.

[1] Personal communication with the organizers, 2026; they invited us to document the mapping here.

AraSafe category	ArGuard label	rows
Illegal Activities	Bully	475
Violence or Harm	harm-to-others	297
Privacy, Misinformation	FraudandDeception	185
Other Harmful Content	self-harm	102
Hate Speech	hate_speech	82
Explicit Content	harassment	29
Terrorism/Extremism	(excluded)	0
(no source)	adult_content	0

Table 1: The organizer-confirmed mapping used to re-annotate the 1,170 novel rows from AraSafe, with the resulting count per ArGuard label (recovered from the released gold). The mapping does not preserve the training convention: adult_content, the largest familiar class, receives no novel examples, and Explicit Content is routed to harassment.

Ensemble. We fine-tune a pool of Arabic encoders, MARBERTv2 (Abdul-Mageed et al., 2021), AraBERT (Antoun et al., 2020), CAMeLBERT-MSA (Inoue et al., 2021), and AraELECTRA (Antoun et al., 2021), plus the multilingual encoder mmBERT (Marone et al., 2025), in several augmented variants (Arabic–English back-translation (Sennrich et al., 2016), Nemotron synthetic-data augmentation, test-time augmentation, and style-mixing), all with class-weighted loss, and average a greedily selected subset with a character n-gram model, chosen on development with a nested-CV guard. The four members selected for the submitted B2 ensemble are MARBERTv2 (test-time augmentation), AraBERT (Nemotron augmentation), mmBERT (style-mixing), and CAMeLBERT-MSA (style-mixing); with the character n-gram model they reach development macro-F1 0.908. Table 2 instead reports the plainer base-encoder ensemble (0.894) so the resolver and decision-bias contributions appear as separate steps; the augmented members reach a comparable score on their own. A leave-one-out ablation (Appendix F) shows mmBERT and MARBERTv2 carry most of the ensemble, while CAMeLBERT-MSA is redundant on development. For B1 we average the four base encoders and apply a tuned threshold.

Retrieval-augmented resolver (B2). Where the top-two margin is small, we consult a large language model (Claude Sonnet 5, pinned model id `claude-sonnet-5`),[2] giving it the nearest labeled training prompts to the query and instructing it to *copy this dataset's convention* rather than apply its own judgment. It may abstain; we override only confirmed-beneficial pairs under a per-pair cross-validated gate. The same model resolves low-margin B1 rows. For the novel slice, where no training convention applies, we instead use a general-definition judge (Claude Opus 5, `claude-opus-5`) that routes each prompt to a harm domain by its plain meaning. All three prompts are reproduced verbatim in Appendix G.

[2]The guidelines permit external datasets for training (e.g., AraSafe) and the organizers confirmed external models/APIs are permitted.

System	macro-F1
B2 dev: ensemble (4 enc. + TF-IDF)	0.8940
+ retrieval-augmented resolver	0.9031
+ decision bias	**0.9070**
B1 dev: 4-encoder ensemble + threshold	**0.9500**
B2 blind test	**0.7875**
B1 blind test	**0.9839**

Table 2: Development macro-F1 (nested-CV) and blind-test results.

Calibration. We add one robust decision bias for the smallest class and, for B1, a single threshold; a full per-class bias vector overfits and is not used.

5 Experimental Results

Table 2 gives the development ablation ladder and the blind-test outcome. Every development increment survives nested cross-validation. Bootstrapping the development predictions (10,000 resamples) gives 95% intervals of B2 0.9070 $[0.894, 0.919]$ and B1 0.9500 $[0.939, 0.961]$. On the blind test, B1 reaches 0.9839 and B2 0.7875, placing **1st (tied) on B1** among 22 teams and **2nd on B2** among 17 in the official ranking.[3] Our development figures also match the benchmark's own reported MARBERTv2 baseline (binary 0.95, granular 0.90) (Zaghouani et al., 2026); we reach 0.950 and 0.907, so we operate at the dataset's intended ceiling.

What we submitted. For B1 we apply the encoder ensemble with a tuned threshold to the familiar rows and label the structurally-identified novel block *safe*; this gives 0.9839. Its coincidence with AraSafe's safe half is an analysis result, not an input to the submitted labels. For B2 we apply the ensemble and resolver to the familiar rows and a topic-domain large-language-model judge to the novel rows. As Section 6 shows, any novel-slice

[3]Our submissions appear on the official leaderboard under the team name `teamlatticeeval`.

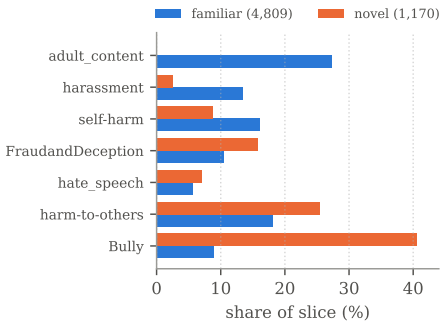


Figure 1: Per-domain share of the novel and familiar B2 test slices. Under the mapping of Table 1, adult_content (the largest familiar class) receives no novel rows while Bully (from Illegal Activities) quadruples. The two columns cannot come from one labeling convention.

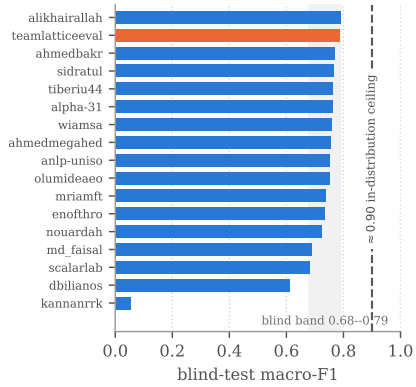


Figure 2: Blind-test macro-F1 for all 17 B2 teams (official ranking). No team exceeds 0.790, and the competitive field sits in a narrow 0.68–0.79 band, far below the ≈0.90 reachable in-distribution (dashed: our development score and the ArabicDialectSafety MARBERTv2 baseline). Our entry (red) is 2nd.

method scores near 0.31, so the pooled 0.7875 is set entirely by the familiar slice.

The gap is one slice. The B2 development score (0.907) and the familiar-slice test performance agree; the drop to 0.7875 is produced entirely by the novel slice, on which every system we built clusters near 0.31 accuracy regardless of architecture or training data. The remainder of the paper explains why.

6 Error Analysis

The two slices use different label conventions. The familiar and novel slices are labeled under conventions that differ by construction (Table 1). Reading the novel labels by content shows the mapping at work: all 21 pornographic prompts are labeled harassment (familiar: mostly adult_content), all 19 suicide prompts are labeled harm-to-others (familiar: mostly self-harm), and 54 of 57 theft prompts are labeled Bully (familiar: mostly FraudandDeception). The label follows the AraSafe source category through the fixed mapping, not the surface meaning a model reads, so a system that classifies content correctly still misses the novel label whenever the mapping and the meaning disagree. The per-domain composition of the two slices makes the divergence visible in aggregate (Figure 1).

Provenance determines the label. Because the mapping is a function of the AraSafe source category, the true category predicts the novel label almost perfectly, whereas any content-based reading of the prompt does not. We saw the corollary during the evaluation phase: a diagnostic submission (not our entry) that recovered the true source category but mapped it to the *semantically nearest* ArGuard domain scored 0.7683, below our honest system, because the gold follows the fixed mapping rather than meaning.

Label-free diagnostics agree. Independently of the organizers' confirmation, three model-only diagnostics already indicated a different label rule rather than difficulty. Fidelity to the training convention is anti-correlated with novel-slice accuracy (Pearson $r=-0.88$ over four systems): our best in-distribution learner is our worst on the slice. Agreement-on-the-line (Baek et al., 2022) over-predicts the novel accuracy by 0.39, the documented footprint of re-labeled data (Sanyal et al., 2024), and the slice shows item-response negative discrimination (Lalor et al., 2016), capable models scoring lower.

The field-wide collapse follows directly. Because the novel slice is scored against every submission and its convention is not inferable during the task, all systems that solve the actual task are penalized alike on that 20% (Figure 2). On the blind test the competitive B2 field compresses into a 0.68–0.79 band: of the 17 teams, the winner reaches only 0.790 and we place 2nd at 0.788, all far below the ≈0.90 macro-F1 that our system and the dataset's own MARBERTv2 baseline reach in-distribution. The collapse is not an annotation-difficulty ceiling; it is the uniform footprint of a single out-of-convention slice, and it caps even the strongest systems.

The honest measure is the familiar slice. Because the novel slice is scored against a convention absent from the task's own data, the meaningful measure of the system is its performance on the familiar slice, macro-F1 0.908, which matches development. The pooled 0.7875 understates every participant by the same out-of-convention 20%. We reproduce the slice's composition from the released gold alone (supplementary script and Table 1); the organizers confirmed the mapping and consider it part of the task design, so the official ranking stands.

Where the familiar errors are. The residual familiar error is the interpersonal cluster: harm-to-others (0.846) and bullying (0.858) hold macro-F1 down (Table 3), where the residue is genuinely dual-target prompts for which a single label is ill-posed, a form of intrinsic label uncertainty (Northcutt et al., 2021). We are near the annotation-agreement ceiling (AraSafe reports 82% fine-grained agreement; we reach 90% on development), and a broad sweep of methods, weak-supervision stacking, learned meta-combiners, prior/label-shift correction, macro-F1 plug-in weighting, and dialect conditioning, is neutral-to-negative in cross-validation (Table 4, appendix).

Leakage audit. In train and development, the data is sorted and blocked by domain, so the identifier alone recovers the domain at 0.9993 macro-F1, and a template-to-label lookup is ≈99.6% pure (Appendix E). We disclosed both to the organizers and used neither. The blind test is properly shuffled (adjacent-block purity matches chance), so the leak is closed there.

7 Conclusion

Our system places 1st (tied) on B1 and 2nd on B2, reaching macro-F1 0.908 on the familiar portion of the B2 test. Its more durable contribution is diagnostic: the B2 test contains a re-annotated novel slice from AraSafe, built under a fixed, organizer-confirmed cross-taxonomy mapping (Table 1) that does not preserve the training label semantics and is not inferable during the task, so it caps every system and compresses the whole field below 0.79. Reporting the familiar-slice score as the honest measure, and aligning such mappings with the training convention, would strengthen future editions of the benchmark.

Limitations

Each configuration is trained once; we report bootstrap intervals over the evaluation set but not seed-variance bands. The resolver can call an external model at inference, though a self-contained fallback (development 0.900) removes this dependence.

Dialect coverage. The training and familiar-test data span Modern Standard Arabic and six regional dialects, but several widely spoken varieties, Gulf, Iraqi, Sudanese, and Yemeni Arabic, are absent, so our reported numbers do not attest to performance on them; the morphological and lexical divergence of these varieties is exactly where a normalization-plus-encoder pipeline is most likely to degrade. We also do not report per-dialect macro-F1, so robustness *across* the dialects that are present is uncharacterized, and a pooled score can hide a weak dialect behind strong Modern Standard Arabic. The novel slice is entirely Modern Standard Arabic, so it stresses convention shift rather than dialect; a deployment audit would need the dialect-stratified evaluation this benchmark does not yet provide.

Our reading of the novel slice rests on the released gold and on the organizer-confirmed mapping of Table 1; the novel-slice analysis relies on an exact-text join to a public corpus, used only diagnostically, never in the submitted system, which partitions the slices from the organizers' files alone.

Ethical Considerations

The data contains harmful prompts; we release no harmful content and report only aggregate results. Synthetic examples used in development train a defensive classifier only and are not released. We disclosed the identifier and template leaks to the organizers and do not exploit them, and we deliberately did not use AraSafe provenance to set submitted labels even where it was accessible, because doing so would exploit test-set leakage rather than demonstrate classification. We shared our observation of the novel-slice labeling with the organizers before publication; they confirmed the mapping in Table 1 and invited us to document it here.

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A Intra-script obfuscation

`normalize_defense` reverses eight of ten disguise vectors: bidirectional controls, zero-width joiners, tatweel, presentation forms, diacritics, alef/ya/hamza variants, non-breaking/thin spaces, and partial homoglyphs; dotless-letter and arabizi-digit substitution are lossy.

B Hyperparameters

Encoders: 3 epochs, batch 32, lr 2×10^{-5}, max length 80, fp16, class-balanced loss, seed 42, single T4. TF-IDF: char n-grams $(2, 5)$, min-df 2, balanced logistic regression. Resolver: margin threshold 0.30, $K{=}12$ neighbours, pinned model. Ensemble members selected by greedy forward selection with a nested-CV guard.

C Per-class B2 results

Domain	P	R	F1
adult_content	0.989	0.988	0.989
hate_speech	0.948	0.941	0.944
fraud	0.898	0.944	0.921
self-harm	0.924	0.910	0.917
bullying	0.851	0.866	0.858
harm-to-others	0.863	0.829	0.846
harassment	0.864	0.885	0.874
macro	0.905	0.909	**0.907**

Table 3: Per-class B2 development results (all rows, the official scoring); the interpersonal cluster bounds the macro average.

D Negative and neutral results

Method (vs. best)	Δ macro-F1
Weak-supervision stack (dev CV)	-0.014
Macro-F1 plug-in weights (dev CV)	-0.003
Dialect prior correction (dev)	-0.001
Learned meta-combiner / SWA (dev CV)	≤ -0.005
Label-shift prior correction (dev)	<0
Semantic provenance mapping (test)	-0.019

Table 4: Representative neutral-to-negative results on the familiar task; the macro-F1 plug-in weighting follows Narasimhan et al. (2015). The last row is the diagnostic provenance mapping of Section 6: mapping by meaning loses because the novel key follows the fixed source-category mapping.

E Leakage details

Identifier-to-domain blocks are contiguous and non-overlapping in train (e.g., adult_content $[0, 6560]$, self-harm $[20162, 24050]$); the identifier alone recovers the domain at 0.9993 development macro-F1, and the template-to-label lookup is $\approx 99.6\%$ pure. Both are absent from the shuffled blind test.

F Ensemble member ablation

Table 5 reports each member's development macro-F1 (all rows, the official metric) and the change from removing it from the four-member submitted B2 ensemble, with the character n-gram model retained throughout. mmBERT and MARBERTv2 carry most of the ensemble; CAMeLBERT-MSA is redundant on development but was kept by the greedy selection, which optimizes a pooled development-plus-test objective. The numbers are reproducible from the cached member predictions.

Member	dev F1	Δ drop
MARBERTv2 (TTA)	0.8881	-0.0059
AraBERT (Nemotron)	0.8815	-0.0016
mmBERT (style-mix)	0.8763	-0.0104
CAMeLBERT-MSA (style-mix)	0.8677	0.0000
TF-IDF char n-gram	0.8511	-0.0023
Full ensemble	**0.9079**	

Table 5: Leave-one-out ablation of the submitted B2 ensemble on development (official all-rows macro-F1). "Δ drop" is the change when that member is removed and the rest are re-averaged.

G LLM component prompts

Each component sends a short system prompt and one user turn, caps the reply at 40 tokens, and parses a single JSON label; a refusal or unparseable reply abstains and keeps the encoder prediction. Model ids are pinned (`claude-sonnet-5`, `claude-opus-5`).

B2 familiar resolver (Claude Sonnet 5). System: "You label Arabic prompts for a defensive moderation shared task. Copy how the similar labeled prompts below were labeled (this dataset's convention). Reply ONLY JSON `{"label":"<A|B|unsure>"}`." The user turn gives the two candidate domains with one-line definitions, the K nearest labeled training prompts to the query with their labels, and then the query.

B1 resolver (Claude Sonnet 5). System: "You label Arabic prompts SAFE or UNSAFE for a defensive content-moderation shared task. Below are the MOST SIMILAR already-labeled prompts from this dataset; copy how THIS dataset labels such cases (its convention), not your own opinion. Reply ONLY JSON: `{"label":"<safe|unsafe|unsure>"}`." The user turn lists the nearest labeled safe and unsafe training prompts, then the query.

Novel-slice judge (Claude Opus 5). System: "You are building a DEFENSIVE Arabic content-moderation classifier for an academic shared task. Every prompt below has already been flagged as unsafe; your only job is to route it to the single harm domain it belongs to. Classify by the prompt's plain, general meaning," followed by one-line general definitions of the seven domains and three routing questions (who is the target: the speaker, one specific person, or a group by identity; for one person, sexual pursuit vs. non-sexual mockery vs.

physical damage; damage vs. profit through deception), ending "Judge the prompt on its own meaning, not on any dataset style. Reply ONLY JSON `{"label":"<one of the seven>"}`." The user turn is the query alone; unlike the resolvers it is shown no training examples, by design.